

TAPAN MANNA

Kolkata, West Bengal, India

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SUMMARY

Final-year B.Tech Computer Science Engineering student specializing in **Java backend development**. Experienced in building scalable **RESTful APIs** and secure backend systems using **Spring Boot**, **Spring Security**, and **Spring Data JPA**. Strong foundation in **Java**, **OOP**, **SQL**, and database-driven application development with hands-on experience in API security, authentication, and backend architecture. Proficient in modern development practices, including Git/GitHub workflows and **API integration**. Seeking a **Java Backend Developer** role to contribute to scalable software solutions.

EDUCATION

Sister Nivedita University

Bachelor of Technology – Computer Science and Engineering

Kolkata, India

Aug 2022 – May 2026

TECHNICAL SKILLS

- **Programming Languages:** Java (Primary), SQL, Python (Basic), C/C++
- **Frameworks & Libraries:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA/Hibernate, RESTful APIs
- **Databases:** MySQL, MongoDB
- **Developer Tools:** Git/GitHub, Maven, Postman, Swagger, Docker (Basic)
- **Core Concepts:** Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, SDLC
- **Soft Skills :** Teamwork, Leadership & Team Management, Communication

EXPERIENCE

Experiential Learner Intern | [View Certificate](#)

April 2025 – July 2025

Remote

Tools: Postman, Burp Suite, Kali Linux

- Performed API security testing using Postman and Burp Suite, identifying and documenting **10+ security vulnerabilities**, improving API reliability, robustness, and secure endpoint validation.
- Applied **OWASP Top 10** security principles while performing validation and verification testing using Kali Linux tools, generating structured test reports and technical documentation for **multiple API endpoints**.

PROJECTS

Personal Finance Tracker | [GitHub](#)

January 2026 – February 2026

- Architected a layered backend architecture (Controller–Service–Repository) using Java, Spring Boot, and Spring Data JPA to manage transactions, budgeting, and real-time expense tracking with persistent storage in MySQL.
- Established JWT-based authentication and role-based authorization using Spring Security to secure REST APIs with stateless session management.
- Developed 15+ RESTful APIs for transaction processing, budgeting, and financial analytics including income–expense summaries, monthly spending trends, and category-wise insights.
- Enhanced application functionality by integrating budget alert notifications, email verification using JavaMailSender, automated Excel report generation with Apache POI, scheduled background jobs using Spring Scheduler, and secure profile image uploads via Cloudinary.

Tech Stack: Java, Spring Boot, Spring Security, Spring Data JPA/Hibernate, REST APIs, MySQL, Cloudinary

AI-Powered Smart Cafe Management System | [GitHub](#) September 2025 – November 2025

- Constructed a scalable backend integrating Spring Boot (Java) with FastAPI (Python) to power AI-driven analytics and REST APIs for cafe operations including products, orders, billing, and user management.
- Engineered machine learning pipelines for sales prediction, product recommendation, peak hour detection, stock forecasting, and business insights using Scikit-Learn, Apriori algorithm, and Facebook Prophet.
- Designed a high-performance unified analytics dashboard API consolidating multiple AI services into a single endpoint and implemented a real-time recommendation engine using Apriori algorithm to analyze transaction datasets.
- Built a time-series forecasting model using Facebook Prophet to predict future sales demand from historical transaction data while strengthening backend security with JWT-based authentication and role-based authorization using Spring Security and optimizing MySQL queries and data pipelines for analytics workloads.

Tech Stack: Java, Spring Boot, Spring Security, FastAPI, Scikit-Learn, ML, Python, Prophet, MySQL, REST APIs

ACHIEVEMENTS & CERTIFICATIONS

- **5-star Java Badge (HackerRank)** – Earned top-level badge in Java programming. [Link](#) **March 2024**
- **Amazon Junior Software Developer (Amazon)** – Accomplished training in backend development and Java fundamentals. [View Certificate](#) **January 2026**
- **Data Structures & Backend with Java (Board Infinity)** – Gained hands-on proficiency in Spring Boot, JDBC, JPA, Hibernate, and RESTful APIs. [View Certificate](#) **June 2025**